

Event Density — Itemized Theory: The Tiered-Claims Ledger

Companion to A Unified Framework for Physics. 439 claims across eight sheets. September 2026.

The report makes the argument. This is the document that holds it to account.

Every load-bearing claim in the Event Density corpus appears here once, with four things attached: what tier it holds, what it rests on, what would kill it, and when it was last checked. Where those disagree with the papers, this file is the one that gets corrected.

The tier grammar is strict, and the strictness is the point. Derived means forced from the thirteen primitives with no paper-specific postulate — nothing else qualifies. Grounded means form-forced given a stated input or a declared postulate, and names it. D-via-I means the form is derived and the numbers are inherited. Inherited names where a value came from. Open states what would close it.

So the headline number is not the one a promotional document would choose: 23 rows of 439 are Derived. 107 are Grounded, 55 Postulated, 51 Open, 41 Selected/Inherited. Beneath the thirteen primitives sit 175 declared postulates, and the corpus's own audit of that count concluded there is no single honest figure — only a ladder, depending on what you count.

What this buys is the ability to be wrong in public. A claim tiered Grounded tells a reader exactly which postulate to attack. A claim tiered Inherited is not pretending to be a prediction. The 51 Open rows are gaps the program declared before anyone else found them — and several rows here record results that moved against the theory: a flagship prediction downgraded to untested when a systematics critique checked out; a repair carried for two days that turned out to invert the problem it was meant to fix; a mass window that lost half its range to an experiment.

Where to start. Core Predictions — the falsifiable bets, and the shortest route to a real objection. Then Ledger w Claims — the full 439, sortable by tier, folder, or what a row rests on. Tier Key defines the grammar; Postulate Flags lists what the postulated rows actually assume.

What this is not. Not peer-reviewed. No distinctive prediction is confirmed. Two structural questions are open, one prerequisite is unproven, and the dimensionless constants are inherited by design and by argument. The verification dates are uneven — 175 of 439 rows carry one — and the corpus's own triage found that roughly 81 of the undated remainder are load-bearing enough to matter; the rest are pointers and superseded entries.

Read it adversarially. It was built to be read that way.

Event Density Theory

Primitives	P01	Existence layer (the foundational relational existence; below it, no further substrate structure)	Primitive
Primitives	P02	Participation as the primitive relation (chains participate — derived composite structures)	Primitive
Primitives	P03	Channel + locus indexing; spatial homogeneity (bundles three commitments, splittable)	Primitive
Primitives	P04	Bandwidth — a non-negative additive scalar	Primitive
Primitives	P05	Polarity-transport between adjacent loci	Primitive
Primitives	P06	Spatial dimension $D = 3+1$	Primitive
Primitives	P07	Channel structure as an ontological primitive	Primitive
Primitives	P08	Substrate scale I_{ED} (= I_P empirically)	Primitive
Primitives	P09	$U(1)$ -valued polarity	Primitive
Primitives	P10	Rule-type primitive (licenses V5 as a P10 kernel rule-type)	Primitive
Primitives	P11	Commitment — irreversible, with environmental phase-randomization (THE arrow primitive)	Primitive
Primitives	P12	Stability landscape $\Sigma = Coh - Str - Grad$	Primitive
Primitives	P13	Time homogeneity; primitive event-discreteness (NOT 'the tick')	Primitive
Constants-inherited	Paper_012 (P-RB-1)	c — the rate of becoming (tick-to-hop conversion)	Constant
Constants-inherited	Paper_027	$G = c^3 \cdot I_P^2 / \hbar$ — Newton's constant	Constant*
Constants-inherited	Paper_029	$a_0 = cH_0/(2\pi)$ — the MOND transition scale. **2026-09-04: the $1/(2\pi)$ is NOT established as forced** — it cancels in Paper_028 §6.3 / Paper_029 §5.1's own displayed algebra and is reinstated by a normalization fixed nowhere. Re-tiered Postulated/disputed (gravity Staleness #10). UNAFFECTED: $a_0 \sim cH_0$, now reached FOUR independent ways, and $a_0(z) = cH(z)/(2\pi)$ exponent 1	Constant* — SCALE ok, COEFFICIENT DISPUTED 2026-09-04
Constants-inherited	Paper_038_5	Λ — the cosmological constant	Constant
Constants-inherited	Paper_087	$\hbar$ — the action of one commitment (the grain of memory)	Constant
Constants-inherited	Paper_087 (P08) / 089	$I_P = I_{ED}$ — the grain (size of one hop)	Constant
Constants-inherited	cosmology	H_0 — the Hubble rate	Constant
Constants-inherited	matter sector	Particle masses	Constant
Constants-inherited	matter sector	Gauge couplings g	Constant
Constants-inherited	matter sector	$\alpha \sim 1/137$ (fine-structure); mass ratios	Constant

Cosmology	SCBU	a0 & xi_canonical FORCED to share one substrate-cosmology boundary origin. **CHECKED 2026-09-04: survives — the shared object is R_H, not the coefficient; the 1/(2pi) dispute does not reach it, and a0 ~ cH0 gained a fourth route the same day. Tier unchanged. Residuals: xi_canonical is canon-INTERNAL and memory-sourced (Foundations staleness #4), and SCBU step 12 leaves xi_canonical (H0) OPEN — so 'FORCED' is conditional on P-Substrate-Cosmology-Unified. See Core Theory row 24.**	Postulated
Entanglement	063	E-1 factorizable $\Psi^{AB} = \Psi^A \times \Psi^B$ & the tensor-product structure	Postulated
Entanglement	064	V5-bilocal has Schmidt rank>1 generically	Postulated
Entanglement	065	Monogamy budget cap; measure saturates the budget	Postulated
Entanglement	069	ED sits at Tsirelson; PR-boxes excluded	Postulated
Gravity	035	Six acoustic-metric guardrails (C1-C6)	Postulated
Gravity	QuadraticStrain	P-Quadratic-Strain (Model C)	Postulated
QFT	019	P-YM-Action-Coarse-Graining	Postulated
QFT	020	P-OS-Reflection-Positivity (OS3)	Postulated
QFT	021	P-Profile-Rescaling (YM continuum survival)	Postulated
QM-kinematics	002	Tensor-product composition $\Psi^{AB} = \Psi^A \times \Psi^B$	Postulated
QM-kinematics	012	P-RB-1 local rate of becoming; substrate-c constancy	Postulated
Relativistic-QM	114	P-Gauge (paper-specific postulate)	Postulated
Substrate-evaluation	QuantumDarwinism	P-QD-LiveWeight (uncommitted branch-basis correlation consumes pairwise V5 weight)	Postulated
Substrate-evaluation	V5AttractiveSign/Relational Tick/MassWithoutMass	V5 existence as a cross-chain kernel	Postulated

Event Density Claims

Black-hole	039	Horizon = $b \rightarrow 0$ decoupling surface (Γ_{cross} collapse)	D-via-I / Form-forced
Black-hole	040	Trans-Planckian resolution (modes terminate at w_c)	D-via-I / Form-forced
Black-hole	041	Planck-mass remnant $M^* = c \cdot I_P$ (Scenario C)	D-via-I / Form-forced
Black-hole	042	No-singularity (C_{cum} bounded, $r \geq I_P$)	D-via-I / Form-forced
Black-hole	043	Area-law FORM $S \propto A$	D-via-I / Form-forced
Black-hole	044	BHPT scattering / Regge-Wheeler form	D-via-I / Form-forced
Black-hole	045	Helicity preserved (Schwarzschild) / amplified (Kerr)	D-via-I / Form-forced

Black-hole	046	Kerr twist $g_{t\phi}$	D-via-I / Form-forced
Black-hole	047	Hawking temperature $T_H = \kappa/2\pi$ via V5 KMS. [2026-09-04: this 2π is NOT the disputed one. It comes from near-horizon Euclidean smoothness (no conical defect) in Paper_BH_Thermal2Pi and does not cancel. The a_0 dispute (row 17) is about the azimuthal-Fourier factor in Paper_029. Whether they are the SAME 2π is exactly Route 2 of research target #18, and is open]	D-via-I / Form-forced
Black-hole	047	Non-thermal Hawking correction $[1 - c_1(w/w_c)^2]$	D-via-I / Form-forced
Black-hole	048	Scenario C forced (A, B excluded)	D-via-I / Form-forced
Black-hole	049	PBH relic abundance $\Omega_{\text{PBH}} \propto n_{\text{PBH}} \cdot M^*$	D-via-I / Form-forced
Black-hole	050	Page-curve rise + turnover + decline	D-via-I / Form-forced
Dark-sector	RelicLagrangian	Relic $\rho \propto a^{-3}$ from the P11 commitment number	Derived
Dark-sector	RelicLagrangian	Sector is TWO-COMPONENT, not one substance (L_{self} coarse-grains CANONICAL)	Derived
Dark-sector	RelicLagrangian	Pressureless-dust limit $w \rightarrow 0$ (fast-oscillating massive scalar)	Derived
Entanglement	064	Schmidt rank $> 1 \Rightarrow$ non-factorizable	Derived
Entanglement	065	Monogamy (CKW-form inequality)	Derived
Entanglement	066	No-signaling $p(a x,y)=p(a x)$, over-determined by 3 locks	Derived
Entanglement	068	von Neumann entropy form $S = -\text{Tr } \rho \log \rho$	Derived
Foundations	089	Theorem N1 — V1 finite-width (G bounded, decaying, scale l_{ED} ; delta-function and inf-width limits excluded)	Derived
Foundations	089/093	Theorem T18 — V1 strictly retarded support (the kernel-level arrow); advanced-V1 refuted by P11, symmetric-V1 non-constructible	Derived
Foundations	089/093	Wheeler-Feynman engagement: symmetric electrodynamics is mathematically consistent but substrate-operationally non-constructible in ED	Derived
Foundations	090/094	V5 inherits retardation from V1 (via T18); primitive-forward-causality - $>$ phenomenological arrows inherit under DCGT	Derived
Gravity	025	Holographic bound $N(R) \leq 4\pi R^2/l_{\text{ED}}^2$ (area law as edge-count)	Derived
Gravity	026	Newton $a_N = GM/R^2$ from the cumulative-strain reading	Derived
Gravity	032.5	Free chains follow acoustic-metric geodesics	Derived
Gravity	034/036	Deep-MOND $a = \sqrt{a_N a_0}$; flat-background AQUAL field eq	Derived
Gravity	037	a_0 continuum-limit invariant (does not RG-run)	Derived

Gravity	GR-I	Temporal lapse $N^2 \sim b \rightarrow$ weak-field Einstein metric (Schwarzschild branch, not Nordstrom)	Derived
Gravity	GR-I	Emergent $g \sim 1/b$ + factor-of-two light bending	Derived
Gravity	GR-II	Field-equation form $G + \Lambda \cdot g = \kappa \cdot T$	Derived
Gravity	GR-II	Matter conservation $\text{div } T = 0 \Leftrightarrow$ geodesic motion	Derived
Gravity	GR-II	ED gravity is KHRONOMETRIC (2 tensor + 1 khronon); gauge = foliation-preserving diffeos	Derived
Gravity	GR-II	$c_T = c$ structurally (GW170817 as identity); universal (not differential) Lorentz violation	Derived
Gravity	GR-III	$\alpha=1$ (P-Commitment-Linear) FORCED; Nordstrom excluded	Derived
Gravity	GR-III	Khronon scalar speed $c_s = c$	Derived
Gravity	GR-IV	$\alpha_2 = 0$ EXACTLY	Derived
Gravity	GR-IV	$\alpha_1 = -4 \cdot \lambda_{\text{local}}$; commitment FORCED sparse $\Rightarrow \lambda_{\text{local}} \ll 1$ (≥ 70 orders safe)	Derived
Gravity	MetricFromGraph	Reach law $p=1/(d-1) \Rightarrow g \sim 1/b$ in EXACTLY 3D	Derived
Q-compute	054	UR-1.1/1.2/1.3 three unresolvedness statements ($M_{\text{eff}} < M_{\text{cap}}$; $\Gamma_{\text{cross}} > \Gamma_{\text{decoupling}}$; $\Gamma_{\text{commit}} < \Gamma_{\text{indiv}}^{-1}$)	Derived
Q-compute	054/056	Existence of Γ_{commit} , $\Gamma_{\text{indiv}}^{-1} \sim c/l_{\text{ED}}$; $M_{\text{cap}} = (\Gamma_{\text{indiv}}/\Gamma_0)^{1/\gamma}$ as the UR-1.1 threshold	Derived
Q-compute	058b	$\alpha_{\text{topological}} = 0$ (per-locus): the 058 §3.3 saturation never triggers for topological codes	Derived
Q-compute	058b	Syndrome records are budget-free (P11 commitments)	Derived
QFT	016	Continuum gauge-covariant derivative $D_\mu = d_\mu + iA_\mu$	Derived
QFT	017	Lorentz covariance of the continuum free-scalar equation	Derived
QFT	019	Continuum YM equations $D_\mu F = J$ (Euler-Lagrange)	Derived
QFT	021	Spectral-gap FORM $\Delta \sim 1/l_*$ (Yang-Mills mass gap)	Derived
QFT	022	Substrate quotient = physical-state space; BRST = coarse-grained quotient	Derived
QFT	024	Lindblad/GKSL master-equation form	Derived
QFT	MS-II	Gauge group $U(N) = SU(N) \times U(1)$ from channel multiplicity	Derived
QM-kinematics	001	Complex polar carrier $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$	Derived
QM-kinematics	009	Berry phase γ_C = holonomy of the coarse-grained P05 connection	Derived

QM-kinematics	010	Aharonov-Bohm phase from T17 bundle circulation	Derived
QM-kinematics	011	Bloch form $\psi_k = e^{i k x} u_k(x)$ from discrete translation symmetry	Derived
QM-kinematics	012.7	Galilean adjacency/propagation bandwidth-asymmetry	Derived
Relativistic-QM	103	Cl(3,1) frame uniqueness up to similarity	Derived
Relativistic-QM	104	Anyon prohibition in 3+1D	Derived
Substrate-evaluation	A1 (CommonCauseNotChannel)	Controlled channel capacity between distinct regions is EXACTLY zero; boundary = common-cause not channel	Derived
Substrate-evaluation	B4 (ChargeAsTopology)	Winding w in Z conserved + irreversibility-protected; orientation-blindness realizes the skeleton, withholds the field	Derived
Substrate-evaluation	CleanSubstrateVector	The parity-clean substrate is VECTOR for every channel-count ($\det H(k)$ even $\Rightarrow$ winding=0 forall N); parity violation NECESSARILY spontaneous	Derived
Substrate-evaluation	QuantumDarwinism	Committed records cost no V5 budget (accounting theorem) $\Rightarrow$ classical objectivity unbounded	Derived
Constants-inherited	Constants-paper §1	Constants ledger: $c/\hbar/l_P$ present at first events (dimensionful, ~units); $a_0/\Lambda/H_0$ /masses inherited late (dimensionless, the personality)	Grounded
Constants-inherited	Constants-paper §3	$G = c^3 \cdot l_P^2 / \hbar$; G absent from the SM; $[G_d] = m^d$ (units know the dimension count)	Grounded
Constants-inherited	Constants-paper §4	The dimension-signature in $G (l_P^{d-1})$ IS the holographic area-law / reach-law road — NOT a new independent road to 3D	Grounded
Cosmology	Cos-01	Inflation = post-SCBU substrate SATURATION regime; $a(t) = a_0 e^{\{Ht\}}$	Grounded
Cosmology	Cos-05	Dark energy = late-universe SATURATION regime; $w = -1$ (leading order)	Grounded
Cosmology	Cos-05	Λ -smallness (120-OOM) reduces to Route A + Friedmann; naive $\rho_{vac} \sim M_P^4$ STRUCTURALLY ABSENT substrate-side (no infinite mode-tower)	Grounded
Cosmology	DF2/DF4	Low-accel galaxies CONSISTENT WITH (not counterexamples to) derived a_0 , via commitment-density saturation suppression	Grounded
Cosmology	SCBU	a_0 & $\chi_{canonical}$ = two projection MECHANISMS of one boundary $R_H = c/H_0$	Grounded
Dark-sector	DarkSector	Committed NEUTRAL RELIC candidate for clusters/CMB	Grounded
Dark-sector	RelicLagrangian	Field content $\Phi = \sqrt{\rho/m_R} e^{i \theta}$ = coarse-grained neutral-channel participation measure	Grounded
Dark-sector	RelicLagrangian	Khronometric kinetic form L_{kin} (time-along- u + spatial c_{perp})	Grounded

Dynamics	Dyn-01	Saddle-Hessian-signature dynamics = the foundational generator; 3 regime classes (S1 collapse / S3 inflation-DE / S2 horizon-radiation)	Grounded
Dynamics	Dyn-02	Horizon-motion law: RDE($w=1/3$), MDE($w=0$), LDE($w=-1$) via substrate Noether stress-energy + DCGT + Friedmann	Grounded
Dynamics	Dyn-03	Radiation law: EM Larmor + GW quadrupole via Noether flux + DCGT	Grounded
Dynamics	Dyn-04	Gravitational collapse = monotonic Hessian-signature evolution toward all-compression; horizon = closed signature-class boundary	Grounded
Dynamics	Dyn-05	Compact-binary inspiral = two-saddle coupled evolution + quadrupole radiation reaction; Peters timescale	Grounded
Dynamics	GW-00	GW = propagating saddle-Hessian-signature reconfiguration (V1 retarded); energy = local ED-gradient load (no pseudotensor)	Grounded
Dynamics	GW-00	Quadrupole source-amplitude FORM $P_{GW} \sim \langle Q\text{-triple-dot}^2 \rangle$	Grounded
Dynamics	GW-01	BH ringdown = decay of a near-horizon mixed-signature saddle; mode quantization from Hessian eigenstructure	Grounded
Dynamics	GW-02	Stochastic background $\Omega_{GW}(f)$ = cosmologically-integrated NoetherFlux over saddle transitions	Grounded
Entanglement	063	E-2 bilocal cross-chain term Δ_{KL} exists	Grounded
Gravity	032	Six ED-10 weak-field prerequisites fulfilled at form level	Grounded
Gravity	033	Scalar-tensor acoustic-metric field eq (covariantization)	Grounded
Gravity	038.5	$\Lambda = V1$ cosmological-scale integral over R_H ; Λ CONSTANT, $w=-1$ forced, = frozen saturation-floor	Grounded
Gravity	KM-I	The khronon's deep field IS the MOND sector (unifies the two gravity programs)	Grounded
Gravity	KM-I	Lensing tracks the MOND potential, no added vector; no new ghost; PPN-safe	Grounded
Gravity	KM-II	Orthogonality theorem: $\theta=0$ in statics $\Rightarrow$ the regulator family is invisible to every tested result	Grounded
Gravity	KM-II	Clustering dial = the soft spot (one dial sets degeneracy-cure + Cerenkov + clusters/CMB)	Grounded
Gravity	KM-II	One-Lambda thesis: $W0 = -24 \cdot \pi^2 \cdot \Omega_{\Lambda} \sim -162$	Grounded
Q-compute	053	$M_{cap} = \min(N_{bw}, N_{V5}, N_{commit})$ is one substrate object; min-structure forced by simultaneity	Grounded
Q-compute	054	The three UR-1 rate quantities have distinct PRIMARY DRIVERS (basis of A/B/C distinctness)	Grounded
Q-compute	055	A/B/C three-class EXHAUSTIVENESS — every architecture saturates one of the three constituents	Grounded

Q-compute	057	Class-B exponential gap-suppression $\Gamma_{\text{fail}} \sim \Gamma_0 e^{-\xi/\xi_0}$	Grounded
Q-compute	058	Class-C plateau $\Gamma_{\text{fail}} \rightarrow \Gamma_{\text{plateau}} > 0$ at redundancy saturation	Grounded
Q-compute	059	Meta-architectures (EC/DD/reservoir/hybrids) = compositions that shift which constituent binds -- NO fourth class	Grounded
Q-compute	060	M_{crit} unification: the matter-wave Q-C boundary and the qubit walls are one substrate threshold	Grounded
QFT	013	V1 QFT-arc propagator (Kallen-Lehmann retarded form)	Grounded
QFT	014	N1+GR1: V1 finite-width + 2-point function on a curved acoustic background	Grounded
QFT	017	Free scalar Klein-Gordon $(\Box + m^2)\phi = 0$ as the DCGT limit of V1	Grounded
QFT	018	Non-abelian YM potential, field strength, YM equation	Grounded
QFT	MS-II	Substrate is a $U(N)$ lattice gauge theory; escapes Nielsen-Ninomiya	Grounded
QFT	MS-II	Spin-1/2 from relationality (tethered node carries $SU(2) \rightarrow SO(3)$ double cover)	Grounded
QFT	MS-II	Parity violation is non-abelian: NO parity-violating abelian force	Grounded
QFT	MS-II	Chirality γ^5 = the arrow's 'screw'	Grounded
QFT	MS-II	$\{1,2,3\} \Leftrightarrow$ internal amplitude dimension $d=3$ (Welch-bound); no stable $SU(N \geq 4)$	Grounded
QFT	MS-II	Three spatial dimensions because the arrow must hold committed order	Grounded
QM-kinematics	003	Born rule $f_K = c_K ^2$ as the participation-frequency limit	Grounded
QM-kinematics	003.5	$P_K = \sqrt{b_K} \cdot e^{i\pi_K}$ amplitude carrier, complex-valued forced	Grounded
QM-kinematics	004.5	Sesquilinear inner-product form $\langle P, Q \rangle = \sum P_K^* Q_K$ (discrete)	Grounded
QM-kinematics	004.6	Continuum inner-product $\int \Psi^* \Phi \, d\mu$ on $L^2(M, d\mu)$	Grounded
QM-kinematics	005	Projective measurement (projector P_K , collapse, Born prob)	Grounded
QM-kinematics	005.5	Double-slit interference $\rho = \sum P_K ^2$; which-path = V5 injection	Grounded
QM-kinematics	006	Unitary evolution $\rightarrow$ Schrodinger $i\hbar \partial_t \Psi = H \Psi$	Grounded
QM-kinematics	006.5	Schrodinger equation via Stone's theorem	Grounded
QM-kinematics	006.6	Kinetic operator $T = p^2/2m$, the factor-2	Grounded
QM-kinematics	006.7	Schrodinger emergence in the thin-participation limit	Grounded
QM-kinematics	007	Hilbert space = Cauchy completion of the motif algebra	Grounded

QM-kinematics	008	Phase structure $S^1/U(1)$ on rule-types	Grounded
QM-kinematics	008.5	Phase-independence of bandwidth = $U(1)$ gauge invariance	Grounded
QM-kinematics	012.5	Momentum operator $p = -i \hbar \text{grad}$ = translation generator	Grounded
QM-kinematics	012.6	Heisenberg $Dx \cdot Dp \geq \hbar/2$	Grounded
Relativistic-QM	102	Spin-statistics $\eta = (-1)^{2s}$	Grounded
Relativistic-QM	106 (T4)	Dirac equation $(i \gamma^\mu d_\mu - mc/\hbar)\Psi = 0$	Grounded
Relativistic-QM	107	Klein-Gordon equation $(\Box + m^2 c^2/\hbar^2)\Psi = 0$	Grounded
Relativistic-QM	108	Minimal-coupling KG + conserved 4-current	Grounded
Relativistic-QM	109	Lorentz-representation taxonomy	Grounded
Relativistic-QM	111	Primitive-level UV finiteness	Grounded
Relativistic-QM	112 (T10)	Lightlike worldlines for $\sigma = 0$ rule-types	Grounded
Relativistic-QM	113	Mass structural form (Arc-M H1)	Grounded
Relativistic-QM	114 (GRH-D1)	Existence of ≥ 1 massless Case-P gauge rule-type	Grounded
Relativistic-QM	115	Vacuum + particle dual Fock structure ($\sigma = 0$)	Grounded
Relativistic-QM	116	MR-R / MR-P conditional massless slot	Grounded
State-reduction	ChannelGranularity	ED's channel/pointer basis is COARSE ($\gg I_{ED}$), so bound charges are single-channel	Grounded
State-reduction	CollapseRate	$\tau \sim \hbar/E_G$ SCALING from ED's primitives (two branch commitment-clocks $N \sim \sqrt{b}$ the single arrow cannot hold $\rightarrow$ einselection individuates)	Grounded
State-reduction	CollapseRate	Two refinements over the rivals: a DEFINITE reduction (not Diosi's mere decoherence) + a NATIVE regulator for the point-mass divergence	Grounded
State-reduction	CollapseRate	NOT excluded by the 2020 spontaneous-radiation bound (Lindblad jump operators = channel projectors $J = g \cdot \Pi$, FORCED from P11; committed states = exact fixed points $D[\Pi] = 0$; no continuous diffusion)	Grounded
State-reduction	OpenPiece-i	MOND enhancement of E_G (mesoscopic masses sub- a_0 in self-gravity $\rightarrow$ naive $\sim 5\text{-}50x$) is CLOSED by ED's own External Field Effect (Paper_029 horizon dipole sourced by total accel; Earth Newtonizes to $\sim 3.5e-6 \rightarrow$ ED $\sim$ Penrose-Diosi in the lab)	Grounded
State-reduction	Positioning §3.2/§6	Locality WITHOUT superdeterminism (local commitment + A1 capacity-zero + stochastic-at-commitment; no hidden variable)	Grounded
Substrate-evaluation	A2 (GrownNotInstalled)	A grown decoupling boundary reproduces exact channel-zero	Grounded

Substrate-evaluation	B4/Maxwell	Integral Gauss law circulation= $2 \cdot \pi \cdot w$ (loop-independent); the Maxwell action is LATENT in the coherence deficit $(\sin^2(d\phi/2) \sim 1/4 (\text{grad } \phi)^2)$	Grounded
Substrate-evaluation	CleanSubstrateVector	$\gamma^5 = \text{arrow} \times \text{spontaneous spatial orientation}$; the vector clean substrate is automatically anomaly-free	Grounded
Substrate-evaluation	GaugeStructure	Gauge group $U(N) = SU(N) \times U(1)/Z_N$ form; F^2 action; mass-gap MECHANISM [A,A] lifts flat direction iff non-abelian; single $U(1)_Y$	Grounded
Substrate-evaluation	Gleason keystone	Channel orthogonality REDUCES to channel-distinctness given the representation; einselection primitive; Born non-contextuality; C selected	Grounded
Substrate-evaluation	V5AttractiveSign/PhaseCoherence	V5's coupling IS the coherence content of the cross-chain superposition $(\sqrt{b_A b_B} \cos(d\phi - \text{int } A))$; MOND cross-term constructive (P12-Coh rewards alignment)	Grounded
Substrate-evaluation	V5UnifiedBudget/Quantum Darwinism	$\rho_{\text{loc}} = 1$: monogamy budget = Class-C plateau budget (one W_{max} spent two ways); per-locus live cap $N_{\text{live}} \leq W_{\text{max}}/w_{\text{bar}}$	Grounded
Black-hole	HorizonTiling	Horizon tiles ~ 1 bit/Planck cell (3 counts: holo=1, frozen 0.78, edge 0.88)	Measured
Gravity	GR-III	Dynamical rule $b \cdot \text{grad}^2 b - \kappa \cdot \rho$: steady state $\text{grad}^2 b \sim \rho$ (0.999); $r_s \sim M$; emergent frozen $b > 0$ horizon; area law $S \sim A$	Measured
Gravity	GR-III	Hawking scaling $\kappa = 1/(2 r_s) \sim 1/r_h$	Measured
Gravity	QuadraticStrain	Phase-coherence constructive sign supplied (P12-Coh rewards alignment)	Measured
QFT	MS-II	Linking held in 3D, erasable in 4D	Measured
QFT	MS-II	Retarded ('arrow') hopping produces net chirality signatures	Measured
Substrate-evaluation	AreaLaw/V5UnifiedBudget	Straddling cross-chain (V5) edge count scales as AREA ($r^{2.02}$), reach-independent; holographic entropy = the edge count ($\sim 0.88/\text{Planck cell}$)	Measured
Substrate-evaluation	KineticLatticeGas/CanonicalPDE	Certified continuum = kinetic lattice-gas (BALLISTIC), NOT a diffusion PDE; and does NOT generate the UDM mobility law ($\beta \sim 0$ not 2)	Measured
Substrate-evaluation	KnotsNotCrystals/RelationaITick	No spontaneous long-range correlation-order in any sector; persistence is rhythmic (V5 binds local proper time, preserving time dilation)	Measured
Substrate-evaluation	MassWithoutMass	Binding confines massless fronts into a sub-luminal inertial composite (mass without mass); mass(binding) != time-dilation (commit-rate)	Measured
Substrate-evaluation	Maxwell	The coherence-weighted continuum limit IS Coulomb, but the DETERMINATE substrate does NOT cast the smooth field (emergent shadow)	Measured

Substrate-evaluation	PhaseCoherence/V5AttractiveSign	P12 coherence rewards alignment but stays FINITE-REACH (no crystal) 2D+3D; V5's attractive local-lock from P12's positive sign (flip destroys it)	Measured
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